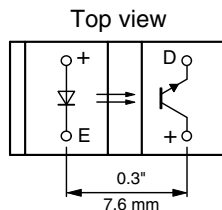


Transmissive Optical Sensor with Phototransistor Output



19180_4



19180_3

DESCRIPTION

The TCST2103, TCST2202, and TCST2300 are transmissive sensors that include an infrared emitter and phototransistor, located face-to-face on the optical axes in a leaded package which blocks visible light. These part numbers include options for aperture width.

FEATURES

- Package type: leaded
- Detector type: phototransistor
- Dimensions (L x W x H in mm): 24.5 x 6.3 x 10.8
- Gap (in mm): 3.1
- Typical output current under test: $I_C = 4$ mA (TCST2103)
- Typical output current under test: $I_C = 2$ mA (TCST2202)
- Typical output current under test: $I_C = 0.5$ mA (TCST2300)
- Daylight blocking filter
- Emitter wavelength: 950 nm
- Lead (Pb)-free soldering released
- Compliant to RoHS directive 2002/95/EC and in accordance to WEEE 2002/96/EC



RoHS
COMPLIANT

APPLICATIONS

- Optical switch
- Photo interrupter
- Counter
- Encoder

PRODUCT SUMMARY

PART NUMBER	GAP WIDTH (mm)	APERTURE WIDTH (mm)	TYPICAL OUTPUT CURRENT UNDER TEST ⁽¹⁾ (mA)	DAYLIGHT BLOCKING FILTER INTEGRATED
TCST2103	3.1	1	4	Yes
TCST2202	3.1	0.5	2	Yes
TCST2300	3.1	0.25	0.5	Yes

Note

⁽¹⁾ Conditions like in table basic characteristics/coupler

ORDERING INFORMATION

ORDERING CODE	PACKAGING	VOLUME ⁽¹⁾	REMARKS
TCST2103	Tube	MOQ: 1020 pcs, 85 pcs/tube	With mounting flange
TCST2202	Tube	MOQ: 1020 pcs, 85 pcs/tube	With mounting flange
TCST2300	Tube	MOQ: 1020 pcs, 85 pcs/tube	With mounting flange

Note

⁽¹⁾ MOQ: minimum order quantity

ABSOLUTE MAXIMUM RATINGS ⁽¹⁾

PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
COUPLER				
Total power dissipation	$T_{amb} \leq 25^\circ\text{C}$	P_{tot}	250	mW
Ambient temperature range		T_{amb}	- 55 to + 85	$^\circ\text{C}$
Storage temperature range		T_{stg}	- 55 to + 100	$^\circ\text{C}$
Soldering temperature	Distance to package: 2 mm; $t \leq 5$ s	T_{sd}	260	$^\circ\text{C}$

ABSOLUTE MAXIMUM RATINGS ⁽¹⁾				
PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
INPUT (EMITTER)				
Reverse voltage		V_R	6	V
Forward current		I_F	60	mA
Forward surge current	$t_p \leq 10 \mu s$	I_{FSM}	3	A
Power dissipation	$T_{amb} \leq 25^\circ C$	P_V	100	mW
Junction temperature		T_j	100	$^\circ C$
OUTPUT (DETECTOR)				
Collector emitter voltage		V_{CEO}	70	V
Emitter collector voltage		V_{ECO}	7	V
Collector peak current	$t_p/T = 0.5, t_p \leq 10 ms$	I_{CM}	200	mA
Power dissipation	$T_{amb} \leq 25^\circ C$	P_V	150	mW
Junction temperature		T_j	100	$^\circ C$

Note

⁽¹⁾ $T_{amb} = 25^\circ C$, unless otherwise specified

ABSOLUTE MAXIMUM RATINGS

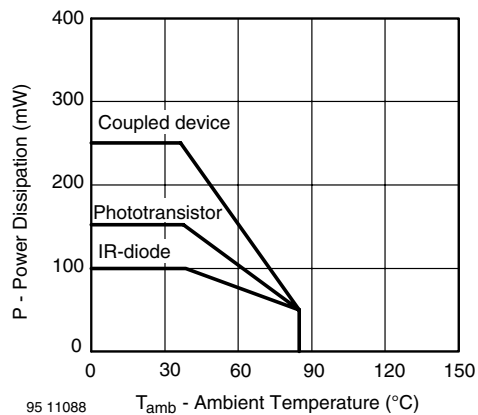


Fig. 1 - Power Dissipation Limit vs. Ambient Temperature

BASIC CHARACTERISTICS ⁽¹⁾							
PARAMETER	TEST CONDITION	PART	SYMBOL	MIN.	TYP.	MAX.	UNIT
COUPLER							
Current transfer ratio	$V_{CE} = 5 V, I_F = 20 mA$	TCST2103	CTR	10	20		%
		TCST2202	CTR	5	10		%
		TCST2300	CTR	1.25	2.5		%
Collector current	$V_{CE} = 5 V, I_F = 20 mA$	TCST2103	I_C	2	4		mA
		TCST2202	I_C	1	2		mA
		TCST2300	I_C	0.25	0.5		mA
Collector emitter saturation voltage	$I_F = 20 mA, I_C = 1 mA$	TCST2103	V_{CEsat}			0.4	V
	$I_F = 20 mA, I_C = 0.5 mA$	TCST2202	V_{CEsat}			0.4	V
	$I_F = 20 mA, I_C = 0.1 mA$	TCST2300	V_{CEsat}			0.4	V
Resolution, path of the shutter crossing the radiant sensitive zone	$I_{Crel} = 10 \% \text{ to } 90 \%$	TCST2103	s		0.6		mm
		TCST2202	s		0.4		mm
		TCST2300	s		0.2		mm

BASIC CHARACTERISTICS ⁽¹⁾							
PARAMETER	TEST CONDITION	PART	SYMBOL	MIN.	TYP.	MAX.	UNIT
INPUT (EMITTER)							
Forward voltage	$I_F = 60 \text{ mA}$		V_F		1.25	1.6	V
Junction capacitance	$V_R = 0 \text{ V}, f = 1 \text{ MHz}$		C_j		50		pF
OUTPUT (DETECTOR)							
Collector emitter voltage	$I_C = 1 \text{ mA}$		V_{CEO}	70			V
Emitter collector voltage	$I_E = 10 \text{ } \mu\text{A}$		V_{ECO}	7			V
Collector dark current	$V_{CE} = 25 \text{ V}, I_F = 0 \text{ A}, E = 0 \text{ lx}$		I_{CEO}			100	nA
SWITCHING CHARACTERISTICS							
Turn-on time	$I_C = 2 \text{ mA}, V_S = 5 \text{ V}, R_L = 100 \text{ } \Omega$ (see figure 2)		t_{on}		10		μs
Turn-off time	$I_C = 2 \text{ mA}, V_S = 5 \text{ V}, R_L = 100 \text{ } \Omega$ (see figure 2)		t_{off}		8		μs

Note

(1) $T_{amb} = 25 \text{ }^\circ\text{C}$, unless otherwise specified

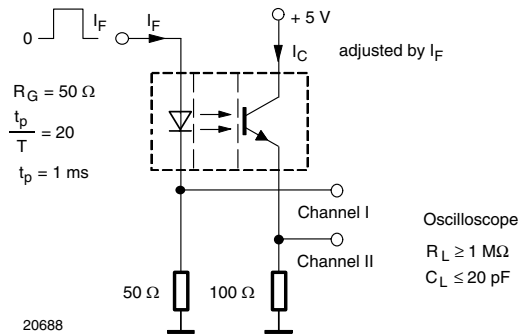


Fig. 2 - Test Circuit for t_{on} and t_{off}

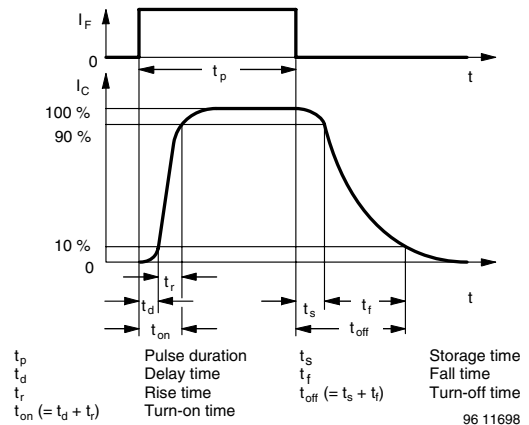


Fig. 3 - Switching Times

BASIC CHARACTERISTICS

$T_{amb} = 25 \text{ }^\circ\text{C}$, unless otherwise specified

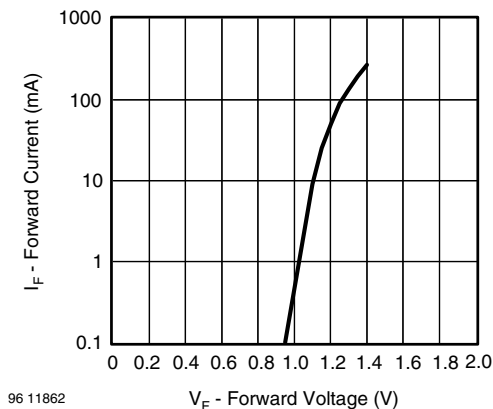


Fig. 4 - Forward Current vs. Forward Voltage

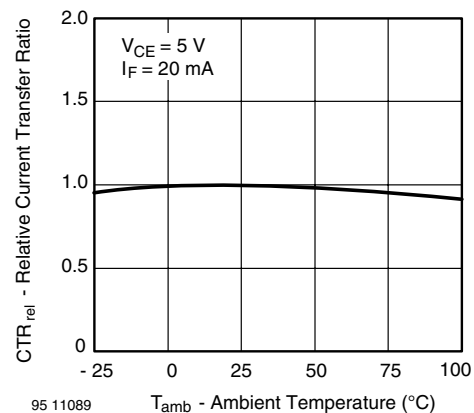


Fig. 5 - Relative Current Transfer Ratio vs. Ambient Temperature

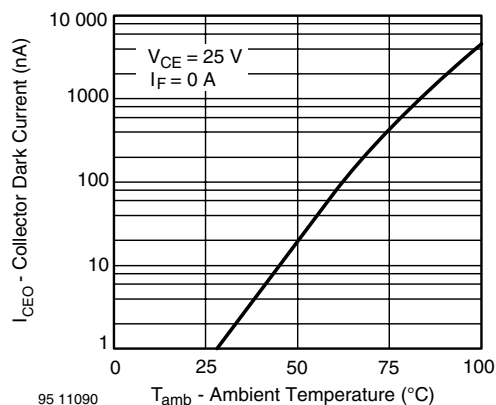


Fig. 6 - Collector Dark Current vs. Ambient Temperature

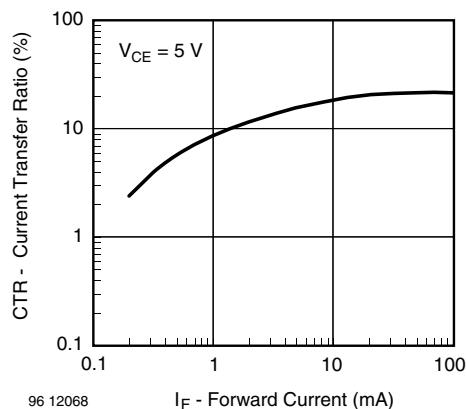


Fig. 9 - Current Transfer Ratio vs. Forward Current

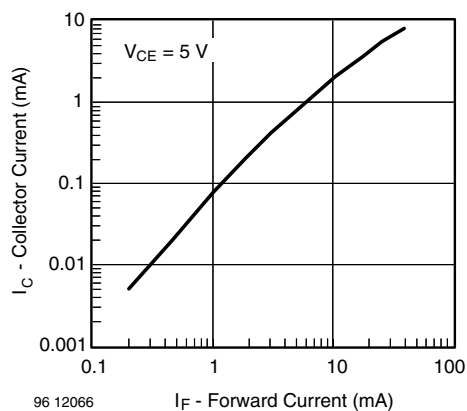


Fig. 7 - Collector Current vs. Forward Current

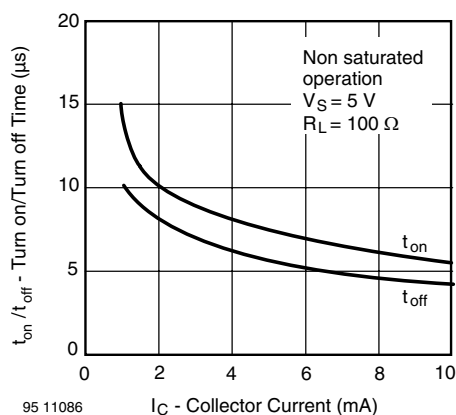


Fig. 10 - Turn-off/Turn-on Time vs. Collector Current

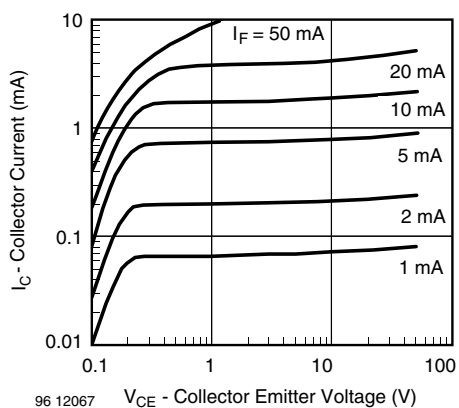


Fig. 8 - Collector Current vs. Collector Emitter Voltage

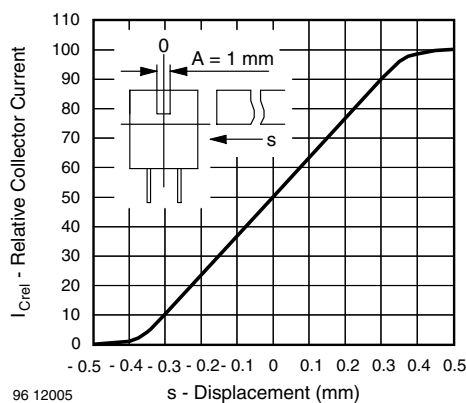


Fig. 11 - Relative Collector Current vs. Displacement

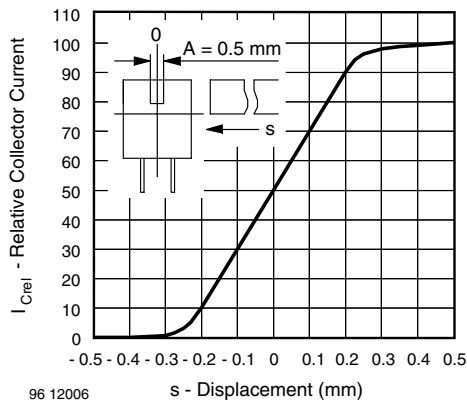


Fig. 12 - Relative Collector Current vs. Displacement

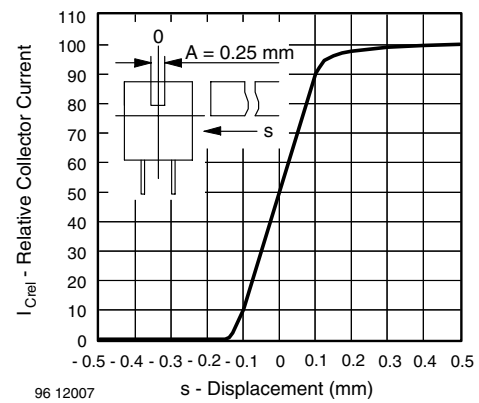
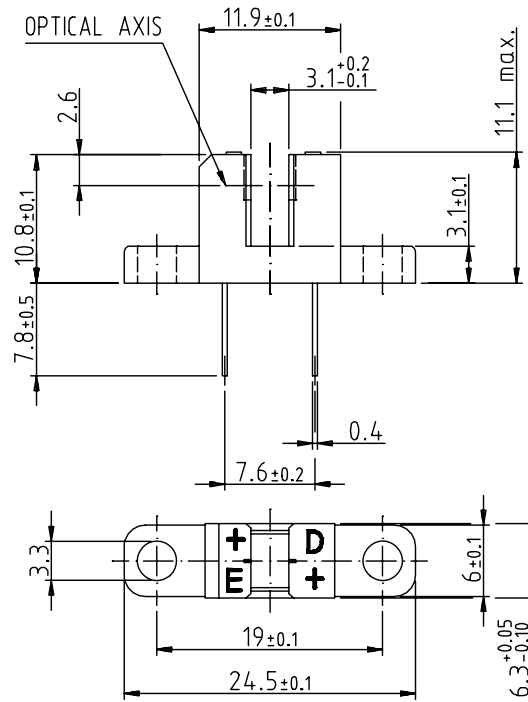
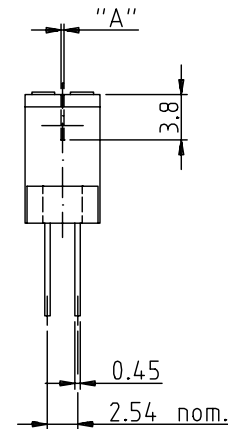


Fig. 13 - Relative Collector Current vs. Displacement

PACKAGE DIMENSIONS in millimeters



Drawing-No.: 6.550-5040.01-4
Issue: 2; 10.11.98
96 12095



technical drawings
according to DIN
specifications

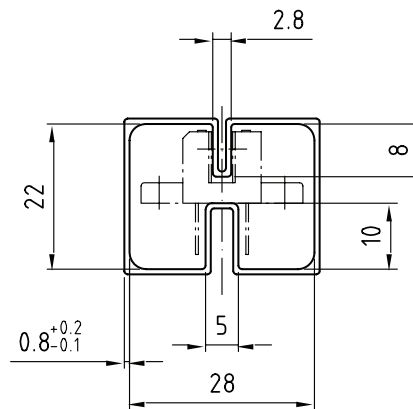
weight: ca. 0.90g

TCST2103, TCST2202, TCST2300

Vishay Semiconductors Transmissive Optical Sensor with
Phototransistor Output



TUBE DIMENSIONS in millimeters



With rubber stopper
Tolerance: $\pm 0.5\text{mm}$
Length: $575 \pm 1\text{mm}$

Drawing-No.: 9.700-5100.01-4

Issue: 1; 25.02.00

20252



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